

■ UZHAVU KAAPPAAN (■■■■■ ■■■■■■■■■■)

Smart Crop Rotation & Farmer Soil Feeding Action Plan
Data Mode: ■ Verified Lab Ingestion (Device: soil-scout-01) · Certified P025 Model

Farmer: Ramesh Kumar | Land Area: 4.5 Acres
Location: Coimbatore, Tamil Nadu | Irrigation: Drip
Plan Ref ID: #UK-P025-FARM-101 | Report Date: 29 Sep 2026, 09:18 PM IST
Live Reading: Active & Calibrated

1. Live Sensor Diagnostic & Soil Fertility Telemetry

■■ Sunlight / Light	■ Soil Moisture	■■ Temperature	■ TDS Minerals	■ Probe Reliability
74% Optimal photoperiod	45% Rhizosphere moisture	31.5 °C Soil probe sensor	420 ppm Conductivity & salts	■ Reliable Soil Scout v2

Nutrient Parameter	Real-Time Value	Target Range	Deficit / Status	Agronomic Impact & Crop Response
Nitrogen (N)	42.0 kg/ha	80 - 160 kg/ha	-38.0 kg/ha (Deficit)	Restricts leaf canopy; urgent pulse rotation needed
Phosphorus (P)	28.0 kg/ha	30 - 60 kg/ha	-2.0 kg/ha (Slight Deficit)	Affects root elongation & early plant establishment
Potassium (K)	55.0 kg/ha	60 - 120 kg/ha	-5.0 kg/ha (Sub-optimal)	Adequate for disease resistance, cell turgor & pod fill
Soil pH	6.50	6.0 - 7.5	Neutral / Optimal	Governs cation exchange & micronutrient availability
Organic Carbon (OC)	0.52%	0.80 - 1.50%	-0.28% Deficit	Governs moisture holding capacity & soil microbiome

2. Prescribed Soil Feeding & Fertilizer Dosage Schedule (for 4.5 Acres Land)

Feeding Stage	Recommended Inputs	Dose / Acre	Total (4.5 Ac)	Application Method & Agronomic Objective
Basal Land Prep (Prior to Sowing)	FYM / Vermicompost Neem Cake Bio-fertilizer (Rhizobium+PSB)	4.0 Tonnes 100 kg 2 kg + 2 kg	18.0 Tonnes 450 kg 18.0 kg	Incorporate during final ploughing to boost Organic Carbon & inoculate root nodules
Sowing Time (Basal Starter)	DAP (Di-Ammonium Phos.) MOP (Potash) Zinc Sulphate (ZnSO4)	25 kg 15 kg 10 kg	112.5 kg 67.5 kg 45.0 kg	Apply in bands 5cm below seed line for rapid root growth & zinc deficiency correction
Vegetative Growth (25-30 Days)	Urea (Top Dressing)	15 kg (-30% saved)	67.5 kg	Green Gram fixes atmospheric N; reduced synthetic nitrogen saves costs and prevents lodging
Flowering & Pods (45-50 Days)	19:19:19 Soluble NPK Borax (0.2% Boron)	1.0 kg (Foliar) 200 g (Foliar)	4.5 kg 0.90 kg	Early morning foliar spray to prevent flower drop and ensure full pod filling

3. Agronomic Risk Assessment & Recommended Restorative Crop Rotation

Season 1 (Immediate Restoration)	Season 2 (Soil Building)	Season 3 (Biomass & Stabilization)
Green Gram (Top Match: 88.5%) • Crop Family: Legume • Duration: 65-75 Days • Est. Net Profit: Rs. 29,000 / acre • Biological Role: Fixes 35-45 kg N/ha naturally and breaks fungal disease cycles.	Groundnut • Crop Family: Legume (Restorer) • Duration: 105-110 Days • Est. Net Profit: Rs. 41,200 / acre • Biological Role: Deep tap roots rebuild subsoil porosity and build Organic Carbon.	Tomato • Crop Family: Poaceae (Feeder / High Biomass) • Duration: 115-120 Days • Est. Net Profit: Rs. 27,300 / acre • Biological Role: Utilizes restored nitrogen, provides heavy organic stubble, and completes recovery.

4. Multi-Season Financial Returns & Soil Health Recovery Trajectory

Rotation Stage	Crop Cultivated	Cost / Ac	Gross / Ac	Net Profit / Ac	Farm Total (4.5 Ac)	Soil Trajectory
Baseline	Depleted Soil	Rs. 36,000	Rs. 48,000	Rs. 12,000	Rs. 54,000	58 / 100 (Current)
Season 1 (Kharif)	Green Gram	Rs. 12,500	Rs. 41,500	Rs. 29,000	Rs. 130,500	65 / 100 (+7 pts)
Season 2 (Rabi)	Groundnut	Rs. 18,000	Rs. 59,200	Rs. 41,200	Rs. 185,400	72 / 100 (+7 pts)
Season 3 (Zaid)	Tomato	Rs. 14,200	Rs. 41,500	Rs. 27,300	Rs. 122,850	79 / 100 (Restored)
3-Season Total	Restorative Rotation	Rs. 44,700	Rs. 142,200	Rs. 97,500 / ac	Rs. 438,750 Total	79 / 100 (Restored)

■ Mandatory Agronomic Field Guidelines for Ramesh Kumar:

1. **Bio-Inoculation:** Treat Green Gram seeds with *Rhizobium* bio-fertilizer @ 25g/kg seed to maximize natural nodulation.
2. **Input Cost Savings:** Do NOT exceed 15 kg Urea/acre; legume root nodules satisfy nitrogen demand while cutting input expenses.
3. **Crop Residue Retention:** Do not burn crop stubbles. Plough haulms back into soil to lift Organic Carbon from 0.52% towards 0.8%.
4. **Precision IoT Monitoring:** Keep Soil Scout probe clean; observe moisture reading (45%) to trigger drip irrigation at 35%.

Certified by: UZHAVU KAAPPAAN P025 Model

Telemetry: Live Soil Scout Probe #soil-scout-01 · 45,000+ Agricultural Training Records

Authorized Agronomist Extension Officer:

Department of Agriculture & Precision Agronomy, Coimbatore, Tamil Nadu